

Cross-supplier SoH forecasting for second-life LFP cells using a PyBaMM-trained neural operator

Krishna Chaitanya Vaddepally

Turno (Blubble Pvt. Ltd.), Bengaluru, India • krishna.vaddepally@turno.club

Theme: Data-driven Methods • Presentation: Oral (poster acceptable)

Second-life reuse of retired lithium iron phosphate (LFP) cells is limited by the cost of predicting future ageing for each cell. Conventional lifetime testing takes months and empirical models often transfer poorly across supplier-specific LFP cell designs. We present a physics-conditioned neural operator that forecasts the future SoH trajectory of a used LFP cell from a short diagnostic characterisation plus the first 50 operating cycles, over an approximately 500-cycle horizon into the deeper second-life regime.

Workflow (Figure 1). Seven anchor cells from three suppliers (CALB, EVE, REPT) are put through a full R&D characterisation suite (~ 25 days per anchor, one-time cohort cost) and long-term-cycled at 0.25–0.5 C. Beginning-of-life and five degradation parameters (SEI growth, lithium plating, negative-electrode active-material loss) are identified by differential evolution against measured cycling. PyBaMM [1] SPMe on Prada2013 LFP chemistry [2] then synthesises 490 perturbed degradation trajectories around the anchors; SoH-offset augmentation expands this to 14,096 (context, target) training pairs covering the second-life SoH range. A monotonic θ -conditioned Neural ODE [3] with a learned context encoder combines three inference inputs: a supplier-level θ prior from anchor DE fits, DCIR and initial-capacity features from a ~ 2 -hour characterisation, and a $K=50$ -cycle SoH context window from the target cell.

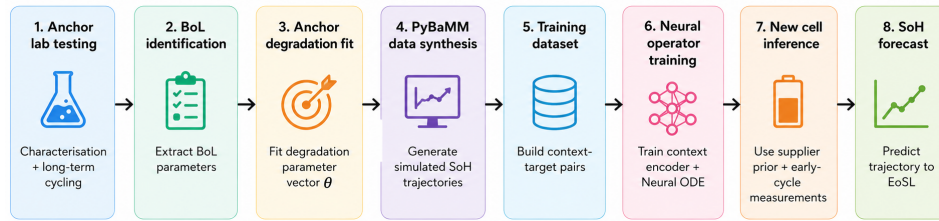


Figure 1: Pipeline: anchor characterisation and DE-fit feed a PyBaMM synthesis corpus; the SciML operator ingests θ , DCIR/capacity features and $K=50$ observed cycles to forecast SoH.

Results. Evaluation on one cell per supplier that was excluded from neural-operator training (CALB_0029, EVE_0003, REPT_0028) gives SoH RMSEs of 0.96, 0.79 and 0.47 percentage points on the experimentally observed forecast region (Figure 2), mean 0.74 pp: preliminary evidence of cross-supplier transfer within the evaluated LFP families. The trained operator forecasts 500 cycles in **18 ms** ($\sim 350\times$ faster than the equivalent PyBaMM SPMe run). The inference input (DCIR + RPT + 50 cycles at 0.5 C) takes ~ 12 days per cell: $\sim 55\%$ **shorter than the full R&D characterisation suite** and $\sim 55\times$ **shorter than cycle-to-EoL testing** (~ 650 days at the pilot anchor θ).

Limitations. Cross-supplier claim rests on three excluded cells; extrapolation beyond the observed cycling window is not yet experimentally verified; baselines against linear/exponential fits and the operator without SoH-offset augmentation are planned next.

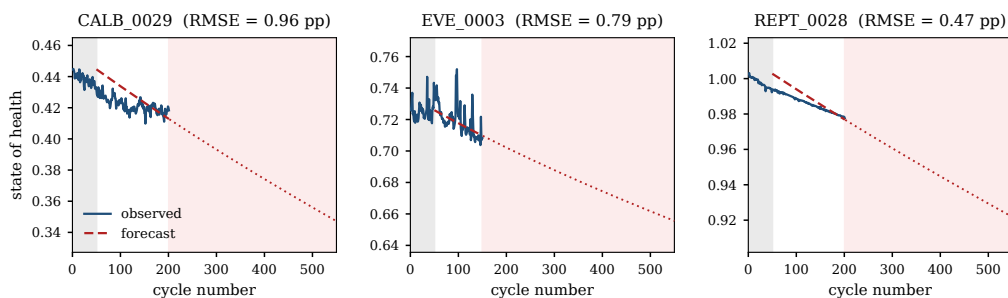


Figure 2: Held-out forecasts (one cell per supplier). Grey band = $K=50$ context; blue = observed; red dashed = forecast; pink band = unverified extrapolation beyond observed window. Per-cell RMSE 0.96, 0.79, 0.47 pp (mean 0.74, std 0.20).